METHOD AND APPARATUS FOR HIGH-RADIX NUMBER REPRESENTATION USING PARAMETRIC SERIES

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Title: METHOD AND APPARATUS FOR HIGH-RADIX NUMBER REPRESENTATION USING PARAMETRIC SERIES

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "Be Water, Be Light," the contents of which are hereby incorporated by reference in their entirety. The method described herein is an improvement upon the high-cardinality parallel data bus described therein.

Abstract

A method and apparatus for representing numbers within a digital computing system. Instead of encoding a number's value directly, the system represents a number as a set of parameters that describe how to generate the number from a known mathematical series. A number is encoded as a "recipe" comprising a series identifier, a position within that

series, and an offset. This parametric representation allows for the efficient encoding of exceptionally large integers. The system also provides for the dedicated encoding of fundamental mathematical constants by reserving specific, high-cardinality states (e.g., specific wavelengths in a non-visible optical spectrum) to represent these constants at a precision limited only by the physical interface, thereby increasing accuracy in scientific and geometric calculations.

Background of the Invention

The representation of numbers in digital computers is fundamentally limited by the bus width of the architecture. A 64-bit computer can only represent integers up to a certain magnitude in a single machine word. Representing larger numbers requires complex software-based multi-word arithmetic, which is inefficient. Furthermore, irrational mathematical constants, such as Pi or Euler's number, are stored as truncated, fixed-precision floating-point approximations. This inherent imprecision can lead to cascading errors in complex calculations, particularly in scientific and geospatial applications.

Summary of the Invention

The present invention provides a novel method for representing numbers in a high-cardinality computing system, enabling the efficient representation of numbers orders of magnitude larger than what is possible with conventional architectures. Instead of transmitting the value of a number directly, the system transmits a "recipe" on how to produce the number. A number, n, is represented by the formula: n = series_id@position + j.

The 'series_id' parameter identifies a known mathematical series (e.g., the series of prime numbers, or a "Wheel" series). The 'position' parameter indicates the index of a specific member within that series. The 'j' parameter is an optional offset. These three parameters are encoded into a single high-cardinality data word. A receiving system can then use this compact recipe to reconstruct the original, potentially massive, number. This changes the communication paradigm from sending numbers to sending instructions on how to generate numbers.

A further aspect of the invention is a method for representing fundamental mathematical constants. The system reserves specific states within the high-cardinality state space to

represent these constants (e.g., Pi, Euler's number, the Golden Ratio). In a preferred embodiment using an optical interface, these constants are assigned to specific wavelengths of light, preferably in the non-visible spectrum to prevent human observation and add a layer of security. This allows the system to compute with these constants at a precision limited only by the physical interface, rather than by a fixed-length binary approximation.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its Federal law enforcement agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting the traditional binary representation of a large number with the parametric series representation of the present invention.

FIG. 2 is a diagram illustrating the dedicated encoding of mathematical constants as reserved states in a high-cardinality optical spectrum.

Detailed Description of the Invention

Referring to **FIG. 1**, a comparison of number representation methods is shown. In the traditional method (100), a large integer (101) is represented as a long string of binary digits (102). The length of this string grows with the magnitude of the number, consuming significant bandwidth. In the present invention (110), the same large integer (101) is represented as a compact set of parameters, or a "recipe" (111). This recipe includes a 'Series ID' (112), a 'Position' (113), and an 'Offset' (114). These parameters are encoded into a single high-cardinality data word (115), which can be transmitted far more efficiently than the full binary representation.

Referring to **FIG. 2**, the method for encoding mathematical constants is shown. A portion of the available state space of a high-cardinality interface, such as an optical spectrum (200), is illustrated. While a range of the spectrum is used for general data transmission (201), specific, reserved states (202) are assigned to represent fundamental mathematical constants. In this embodiment, specific wavelengths in the non-visible light spectrum are used. For example, a specific infrared wavelength (203) is reserved to represent Pi (3.14159...), another (204) represents Euler's number (e), and another (205) represents the Golden Ratio ('phi'). When a computation requires one of these constants, the system can use this dedicated state, providing a representation of the constant at a much higher precision than would be possible with a standard floating-point variable.

Claims

What is claimed is:

1. A method for representing a number in a digital computing system, comprising the steps of:
    - selecting a predetermined mathematical series;
    - identifying a target number's value as a member of said series at a specific position, optionally with an offset;
    - creating a data word that comprises a series identifier for the selected series, the position, and the offset; and
    - transmitting or storing said data word as the representation of the target number.
2. The method of claim 1, wherein the data word is a single high-cardinality data word transmitted over a parallel interface.
3. The method of claim 1, wherein the mathematical series is a series of integers not divisible by a set of small prime numbers (a "Wheel" series).
4. A digital computing system for representing numbers, comprising:
    - an encoding module configured to receive a target number and generate a parametric representation of said number, wherein the representation comprises a series identifier, a position, and an offset; and

- a high-cardinality interface for transmitting or storing said parametric representation as a single data word.

5. A method for representing mathematical constants in a digital computing system with a high-cardinality interface, comprising the step of:

    - permanently reserving at least one discrete state of the high-cardinality interface to represent a specific, fundamental mathematical constant.

6. The method of claim 5, wherein the high-cardinality interface is an optical interface and the discrete state is a specific wavelength of light.

7. The method of claim 6, wherein the specific wavelength of light is in the non-visible spectrum.

8. The method of claim 5, wherein the mathematical constant is one of Pi, Euler's number, or the Golden Ratio.

Drawings

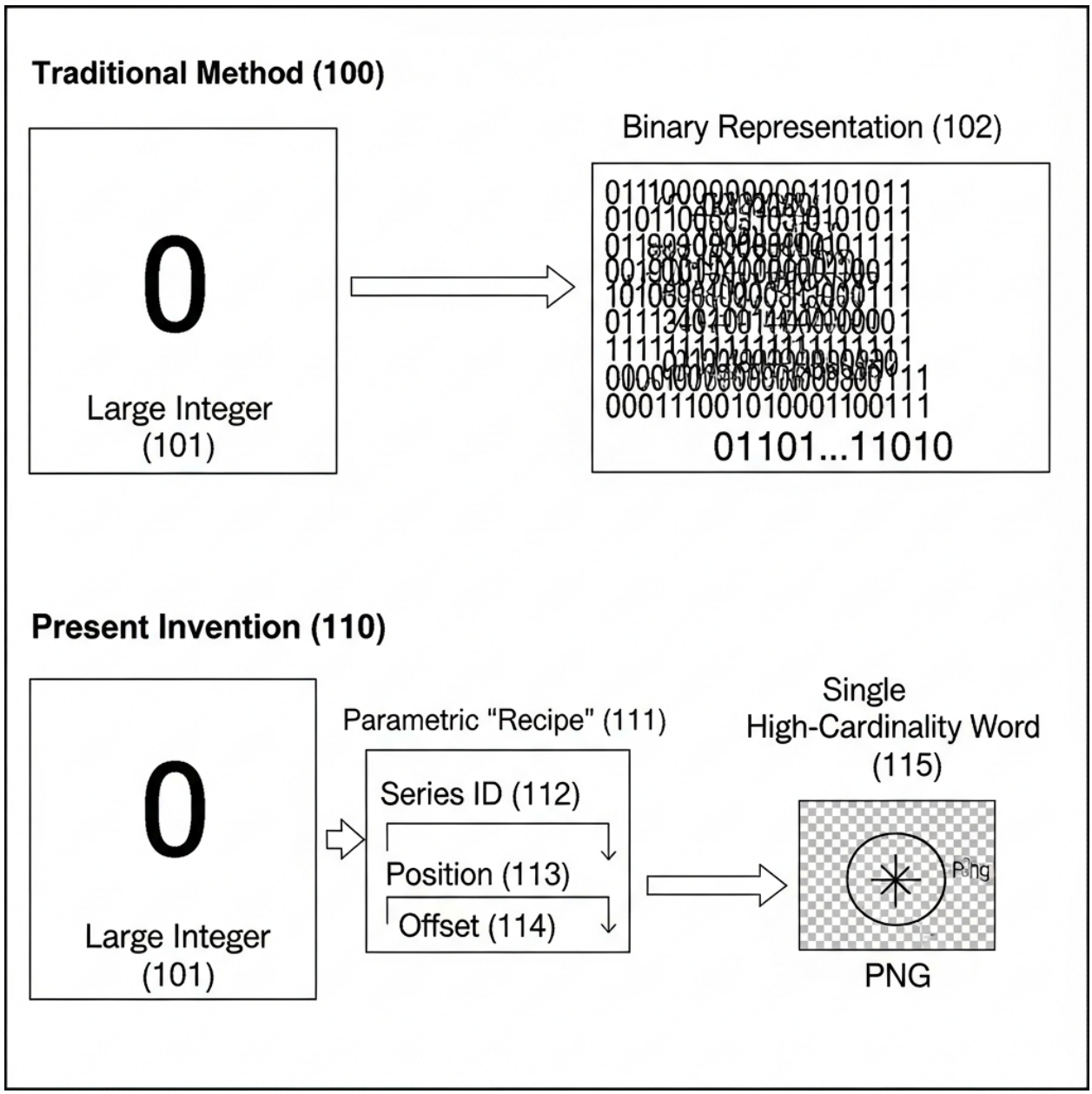


Figure 1: Conceptual diagram contrasting binary vs. parametric representation.

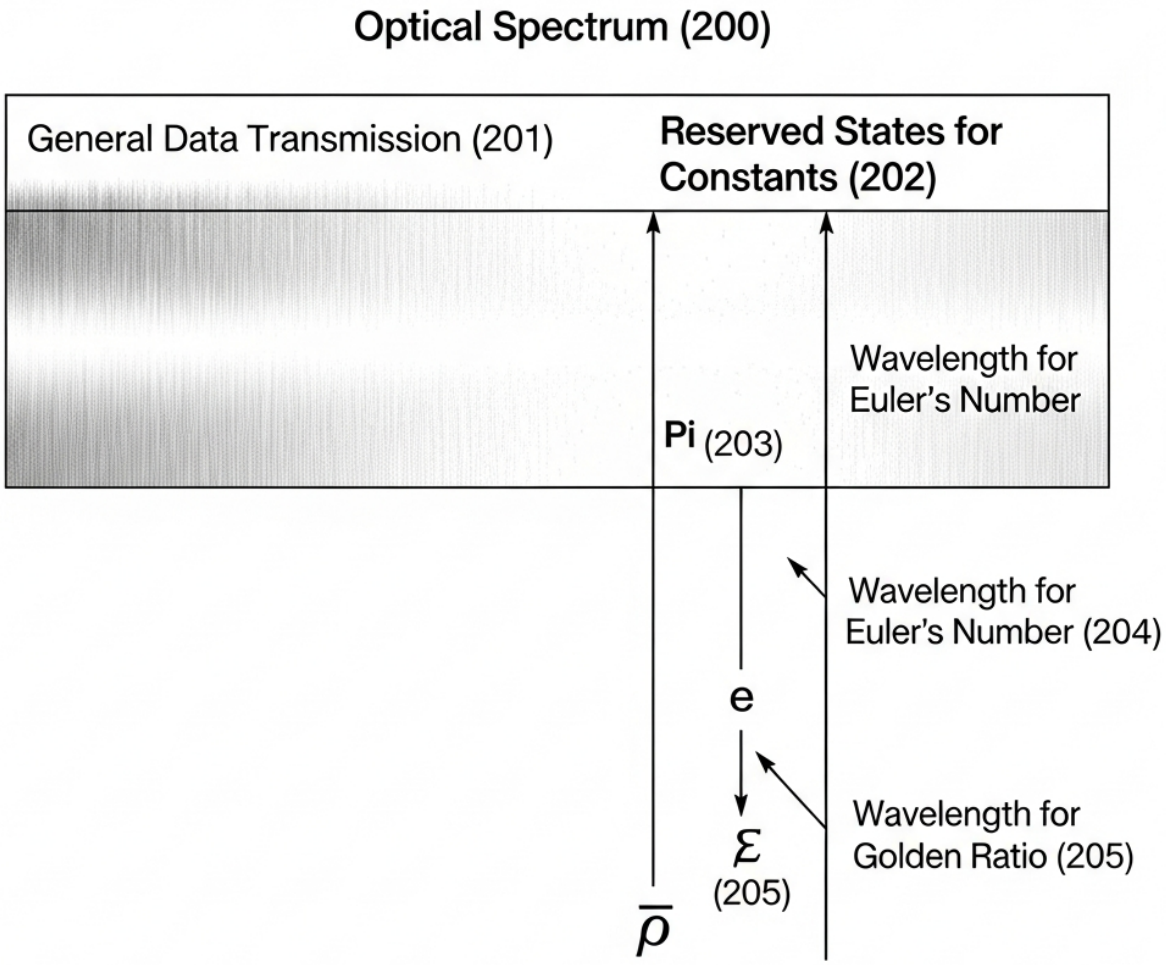


Figure 2: Diagram illustrating dedicated encoding of mathematical constants.